MYEONGWON CHAE

Champaign, IL, USA | aaronchae.github.io

EDUCATION

University of Illinois Urbana-Champaign, IL, USA
- *Ph.D.*, GPA: 3.96/4.0 — Expected May 2029
- *Master of Science in Architectural Studies*, GPA: 4.0/4.0 — Dec 2024

Ajou University, Suwon, Korea
- *Bachelor of Architecture* (Intensive Major Course), GPA: 3.7/4.5 — Feb 2022

PROFESSIONAL EXPERIENCE

Research Assistant — Dec 2023 – Present
The Responsive Architecture Lab, University of Illinois Urbana-Champaign
- Applied machine learning and retrieval-augmented **large language models** to analyze energy consumption patterns and generate structured insights from technical documents for building retrofit decision support.
- Developed data-driven methods to evaluate building energy performance using smart thermostat data, including analysis of HVAC system sizing suitability and operational efficiency.
- Built and calibrated building energy simulation models using **DesignBuilder** and **EnergyPlus** to support performance evaluation and potential energy savings analysis.

Retro-Commissioning Team — May 2024 – Aug 2024
Facilities & Services (F&S), University of Illinois Urbana-Champaign
- Analyzed energy consumption trends in on-campus buildings and handled HVAC system plans.
- Built and calibrated energy models of campus buildings using **eQUEST**.

Teaching Assistant — Aug 2023 – Dec 2024
School of Architecture, University of Illinois Urbana-Champaign
- Instructed 12 graduate students on the use of **DesignBuilder** for building energy simulation.
- Taught 15 undergraduate students and served as the supplemental instructor for weekly design sessions.

Researcher — Feb 2023 – May 2023
Urban & Building Energy (UBE) Lab, Ajou University, Suwon, Korea
- Derived new regional heating and cooling start temperatures from building energy usage data to replace the existing balance point temperature.

Undergraduate Researcher — May 2020 – Aug 2021
Urban & Building Energy (UBE) Lab, Ajou University, Suwon, Korea
- Researched ways to expand and promote the use of the building energy performance information disclosure system.
- Verified energy efficiency diagnosis and retrofit technologies for existing public buildings.

PUBLICATIONS

Journal Articles
- Hye Gi Kim, Hyun Jun Kim, Chae Hwan Jeon, **Myeong Won Chae**, Young Hum Cho, and Sun Sook Kim. "Analysis of Energy Saving Effect and Cost Efficiency of ECMs to Upgrade the Building Energy Code." *Energies*, 13(18), 4955, 2020. doi:10.3390/en13184955

Conference Papers
- **Myeongwon Chae**, You-Jeong Kim, and Yun Kyu Yi. "Data-Driven Assessment of Air-Conditioning System Sizing Using Runtime and Indoor Humidity Performance." *12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings (IAQVEC 2026)*, Los Angeles, CA, USA, May 2026. *E3S Web of Conferences*, 716, 02008. doi:10.1051/e3sconf/202671602008
- **MyeongWon Chae**, Hye-Gi Kim, and S. S. Kim. "Derivation of heating and cooling start temperature by region using building energy use data." *Spring Conference of the Architectural Institute of Korea*, Seoul, Korea, April 2023, pp. 404–405.

Preprints and Accepted Abstracts

- **Myeongwon Chae** and Seonghyuk Son. "From Sound to Occupancy: Privacy-Aware Audio Sensing for Buildings." *arXiv preprint* arXiv:2611.11564, November 2026.
- **Myeongwon Chae** and Yun Kyu Yi. "A multi-agent retrieval-augmented framework for building retrofit decision support with simulation-based validation." *Building Simulation 2027 (BS2027)*, Vienna, Austria, August 29 – September 1, 2027. Abstract accepted.

AWARDS AND HONORS

- **Clarence T Paul Fellowship**, Illinois, USA — Aug 2025
- **JAQ'D Architecture Scholarship**, Illinois, USA — May 2024
- **Fellowship**, Munju Scholarship Foundation, Seoul, Korea — Feb 2021
- **Dean's List**, Ajou University, Suwon, Korea — Feb 2021
- **Grand Prize Award**, An Idea Contest to Revitalize the Space of Smart Green Student Plaza, Ajou University, Suwon, Korea — Oct 2020
- **Encouragement Award**, 2019 Central Urban Regeneration Candidates University Student Architecture Urban Studio, Korean Society of Urban Planning, Seoul, Korea — Nov 2019

MILITARY SERVICE

Sergeant Mar 2016 – Dec 2017

Republic of Korea Army, Inje-gun, Gangwon, Korea

- Completed compulsory military service as an administrator and field engineer; honorably discharged.

EXTRACURRICULAR ACTIVITIES

Korean Student Association (KSA) Dec 2023 – Dec 2024

University of Illinois Urbana-Champaign

- Supported corporate recruiting events, attracted corporate sponsors, and organized academic exchanges between universities in the Midwest.

Student Ambassador (GAIA) May 2014 – Oct 2015

Ajou University, Suwon, Korea

- Carried out briefing sessions on admissions procedures and campus tours.

SKILLS

- **Programming:** Python, R (RStudio)
- **Building energy simulation:** EnergyPlus, DesignBuilder, OpenStudio, eQUEST
- **Design:** SketchUp, Rhino, AutoCAD